HITESH GUPTA

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Coimbatore, India — relocating to Delhi NCR (Gurugram / Noida / New Delhi); open to remote | Notice period: 15 days or less

Software Engineer · Backend · Cloud · Applied AI

3 national public-sector deployments (IOCL · NHAI · Chennai Metro) • 200+ devices live • 1M+ data points / device / week • ~40% faster queries

SUMMARY

Software engineer (B.Tech, ECE) with 1.4 years owning production backend and cloud systems end to end. At Carbelim I deliver for national public-sector clients — the Indian Oil (IOCL) R&D Centre, NHAI and Chennai Metro Rail — writing Python and Node.js services, running AWS infrastructure and real-time data pipelines, and shipping the React/Next.js dashboards above them. Led a 3-engineer team through a CCUS pilot projected to cut 1,000+ kg CO_2/year. Applied AI is part of my daily build loop: LLM APIs, RAG, prompt engineering and agentic workflows for both development and data analysis.

CORE SKILLS

Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL, Bash
Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, MQTT, microservices, event-driven and real-time streaming services, system design
Cloud & DevOps: AWS — EC2, Lambda, S3, DynamoDB, IoT Core, CloudWatch, API Gateway, IAM; Docker, Linux, Git, CI/CD, GitHub Actions, monitoring, alerting, production support
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Supabase, Neon — schema design, query optimization, ETL and data pipelines
Applied AI: LLM APIs, RAG, prompt engineering, agentic workflows, AI-assisted development and analysis (Claude, Codex, Cursor, GitHub Copilot); pandas, NumPy, scikit-learn
Frontend: React.js, Next.js, React Native, Vue.js, Capacitor.js — responsive dashboards, real-time data visualisation, cross-platform apps
Practices: SDLC, Agile/Scrum, code review, release validation, integration and UAT, technical documentation, direct client and stakeholder delivery

EXPERIENCE

Carbelim Private Limited **Jun 2025 – Present**
Software Engineer *Coimbatore, India*

- Own end-to-end software delivery across three national public-sector engagements — backend services, AWS infrastructure, real-time pipelines, dashboards, deployment and production support — working with the CTO, founding team, embedded engineers.
- **IOCL R&D Centre, Faridabad —** Led technical development of a Carbon Capture, Utilization & Storage monitoring system from scratch, directing 3 engineers from PCB bring-up through firmware validation, OTA flows and cloud deployment; built the AWS EC2/Lambda/DynamoDB backend and C/C++ firmware with MQTT telemetry for real-time CO_2 monitoring. Pilot projected to cut 1,000+ kg CO_2/year; nationwide IOCL rollout pending approval.
- **NHAI Aerocity, New Delhi —** Shipped a production environmental-monitoring system on an AWS IoT Core → Lambda → S3 → CloudWatch ingestion pipeline with live dashboards, threshold alerting and edge watchdog recovery; ran on-site installation, field calibration and end-to-end validation through to formal client handoff, achieving unattended always-on operation on a high-traffic public site.
- **Chennai Metro Rail —** Ran a 5-day continuous indoor air-quality audit across 10 stations capturing 9 parameters (CO_2, PM2.5, CH_2O, O_3, CO, TVOC, NO_2, temperature, humidity); built the Python preprocessing, cleaning and multi-sensor data-fusion pipeline, then analysed the multi-location dataset with LLM tooling to surface trends and anomalies into certification-grade reporting.
- **Backend & performance —** Build and operate Node.js/Express and Python (FastAPI) services exposing REST and WebSocket APIs over MQTT ingest; cut MySQL query time ~40% through indexing and query tuning, and designed the architecture and SOPs scaling toward real-time monitoring of 50K+ devices.
- **Dashboards & apps —** Built cross-platform client dashboards and mobile apps in Next.js and React Native on secure Node.js/Express APIs, delivering live telemetry views, historical export, threshold configuration and relay control.
- **Engineering practice —** Git and CI/CD (GitHub Actions) workflows, code review, release validation, integration/UAT and production support across all three engagements; wrote the SOPs and handover documentation each client signed off.

Carbelim UK Ltd **Apr 2024 – Aug 2024**
Embedded & Full-Stack Developer Intern *Coimbatore, India*

- Built an air and water monitoring device (10+ sensors, custom PCB, microcontroller) and its data platform across 8+ AWS services — ingestion, storage and the APIs behind real-time and historical views; the platform now runs on 200+ deployed devices.
- Shipped the cross-platform web and mobile applications (Vue.js, Node.js, Express.js, Capacitor.js) for live visualisation, historical export and relay control; validated with 100+ testers and iterated through to release.

PROJECTS

- **Presently by Ayuvam — presently.ayuvam.com · 2025 —** Built and shipped a full-stack client-deliverables SaaS on Next.js, PostgreSQL (Neon) and Vercel: secure REST APIs, magic-link sharing, role-based approval workflows, nested file organisation and full-text search across client deliverables. Designed the schema, built the API layer and shipped it to production.
- **Smart Environmental Monitoring System · 2023–24 —** End-to-end AWS platform processing 1M+ data points per device per week, from 10+ sensors through real-time ingestion to an API-driven dashboard with relay control.

EDUCATION

National Institute of Technology, Arunachal Pradesh **2021 – 2025**
B.Tech, Electronics & Communication Engineering — 7.00 CGPA

ADDITIONAL

AWS Solutions Architect Associate (SAA-C03) — in preparation. Led a 3-engineer team at one year of experience. Also a freelance 3D artist: 100+ client projects, including a shot aired in a BBC series.